Asymmetric Bonding: A Self-Enforcing Settlement Primitive for N-Party Coordination

Alessandro Daliana*

April 2026

Abstract

We present FIGARO, a settlement mechanism that enables self-enforcing agreements between mutually distrusting parties without arbitrators, timeouts, or governance. The core primitive is *asymmetric bonding*: each party locks collateral equal to twice the transaction value; only the buyer can trigger resolution. Cooperation is thereby the weakly dominant strategy in a one-shot game, and the unique profile surviving iterated elimination of weakly dominated strategies, producing a Nash equilibrium that holds without repeated interaction.

The mechanism scales from two-party exchange to N-party process trees through *progressive collateralization*: each seller bonds against the cumulative value of all upstream work, not merely their own payment. We prove that this preserves the Nash equilibrium at every chain position, with the risk-to-reward ratio increasing geometrically for downstream participants.

All orders within a process resolve *atomically*—all or nothing—inducing a weakest-link subgame among sellers whose peer-enforcement properties we show to obtain under strictly weaker assumptions than the joint-liability mechanisms classically analyzed in the microfinance literature.

This paper is the mechanism-design half of a multi-paper development. The implementation as a Solidity smart contract, with formal verification across multiple methods, is presented in the implementation paper. Implications for organizational economics—specifically, the Coasean shift toward transaction-scoped institutions—are developed in the institutional-economics paper.

Keywords: mechanism design, Nash equilibrium, collateral bonding, multi-party coordination, process trees, peer enforcement

1 Introduction

1.1 The Problem

Economic coordination between strangers requires trust infrastructure. When two parties who have never met wish to exchange value, they face the classic commitment problem: either party may defect after the other has performed. The historical solution is *institutional intermediation*—firms, courts, escrow services, platform operators—each of which introduces costs, delays, jurisdictional dependencies, and power asymmetries.

The problem intensifies in multi-party settings. A food delivery involves a cook, a kitchen operator, an ingredient sourcer, and a courier; a procurement process involves engineers, manufacturers, inspectors, and logistics providers. The standard architectural response is to aggregate

*Corresponding author. `adaliana@figaro.org`

participants into a single legal entity that internalizes the coordination problem and resolves it through hierarchical management.

Blockchain-based smart contracts have produced a rich literature on trustless exchange, but most protocols address financial operations—swaps, lending, derivatives—rather than general coordination. Those that target coordination (e.g., Kleros 2019, Aragon 2017) typically reintroduce the intermediary as an on-chain dispute resolution mechanism: a jury, a governance body, or a timeout that unilaterally releases funds.

1.2 Our Contribution

We present FIGARO, which achieves self-enforcing coordination through a single mechanism: *asymmetric bonding*. The mechanism's contributions are:

(i) **A minimal settlement primitive** with no timeout, no governance, and no escape hatches from the bonded state. We show that this minimality is not a simplification but a *security requirement*: every additional exit path weakens the Nash equilibrium (Theorem 4.7).

(ii) **Progressive collateralization** for N-party process trees. We prove (Theorem 5.3) that cooperation remains weakly dominant at every position in an arbitrary-depth process tree (with strict dominance whenever at least one other player's cooperation is not already foreclosed), and is the unique profile surviving iterated elimination of weakly dominated strategies; the seller's risk-to-reward ratio increases with chain depth.

(iii) **Atomic resolution** as a coordination forcing function. We show that all-or-nothing process settlement induces a weakest-link subgame among sellers, creating endogenous peer pressure without explicit communication or governance.

(iv) **A reduction over joint-liability microfinance.** We prove (Theorem 6.1) that the peer-enforcement outcome common to Ghatak [1999] and Stiglitz [1990] is obtained by the bonded commitment mechanism under strictly weaker assumptions: no repeated interaction, no local information, no exogenous punishment technology, and no joint-liability contracting.

(v) **Self-enforcing DAG topology** without on-chain validation. We prove (Theorem 5.9) that misreporting cumulative value is a strictly dominated strategy, making on-chain parent-relationship validation unnecessary.

The mechanism's implementation as a Solidity smart contract, the formal-verification stack used to audit it, the threat model, and the three-layer enforcement architecture (economic / coordination / evidence) are developed in the companion implementation paper. The implications of a fixed-cost enforcement primitive for the boundary of the firm in the sense of Coase [1937]—and specifically the demotion of the firm from privileged organizational container to one pattern among many when the subordination overlay is no longer enforcement-driven—are developed in the companion institutional-economics paper.

1.3 Paper Organization

Section 2 surveys related work. Section 3 presents the mechanism formally. Section 4 proves the game-theoretic properties. Section 5 extends the model to N-party process trees and DAGs. Section 6 states the reduction over joint-liability microfinance. Section 7 concludes with open theoretical questions.

1.4 Three Tiers: Kernel, Protocol, Runtime

FIGARO as described in this paper is a mathematical object: a set of definitions, an equilibrium, and the theorems connecting them. Across the companion papers, three tiers are distinguished that the present paper uses only once but that subsequent companions invoke frequently:

- **Kernel** — the irreducible settlement primitive formalized here: the `commit` and `resolveProcess` operations, the bond structure, and the equilibrium they induce. Every claim in the present paper is a claim at this tier.
- **Protocol** — the kernel together with ancillary mechanisms that compose with it under a preservation discipline (attestation coordinators, schema registries, auction primitives, operator registries). The composition discipline itself is developed in the companion implementation paper.
- **Runtime** — the protocol together with semantic layers, builder surfaces, user interfaces, and the vertical-specific deployments that instantiate it for a given industry or community. Runtime choices (specific tokens, specific schemas, specific operator communities) are out-of-scope for the present paper.

Claims about "the primitive" that appear in companion papers refer, unless noted otherwise, to the kernel tier. Properties sometimes loosely attributed to the primitive — schema-typed attestations, NFT-anchored credentials, specific token denominations, accepted-token lists — live at the protocol or runtime tiers and are guaranteed only by the additional mechanisms that implement them.

The kernel is ideologically agnostic; the graph is the politics. A consequence of locating the whole of the present paper's argument at the kernel tier is that the kernel admits no commitments about *what* is being coordinated or *under what normative regime*. Currency, jurisdiction, identity type, arbitration forum, role structure, price-discovery mechanism, contribution metric — none of these are in the kernel; all of them are expressed in the *graph* that parties compose on top of it. A Dutch-auction-and-private-operator graph expresses a market-liberal coordination pattern; a cooperative-bonding-and-contribution-weighted-resolution graph expresses a cooperative one; an interest-free risk-sharing graph expresses an Islamic-finance one. The kernel enables each and prefers none. This is not evasion. It is the architectural consequence of placing the enforcement function at TCP/IP altitude: a layer below which nothing remains to factor out, and a layer above which the normative choices become legible as choices rather than as properties of the infrastructure. Claims in the companion papers that bear on ideology or normative regime — the political-economy companion paper on cultural hegemony; the labor-law companion paper on subordination — are claims about what becomes *expressible* in the graph once the kernel is in place, not claims about the kernel itself.

Composability is external-first. A second consequence of the kernel's narrow construction is that the protocol's useful extensions are predominantly *external* rather than internal. The kernel does not include a dispute forum (Kleros, the Singapore International Arbitration Centre, or any court); it does not include a carbon-offset market (Klima DAO, Toucan, Moss); it does not include a prediction market, an insurance pool, a lending facility, a tax-reporting service, an identity provider, a storage layer, or a messaging fabric. Each of these exists as a working external protocol or institution, and an assembly composed at the runtime tier can name any of them in its graph: a forum-clause names the dispute forum the parties agree to; an attestation clause names the offset operator the buyer wants to compose into the process tree; a sub-order edge in the tree carries the buyer's offset purchase to the named operator within the same atomic resolution as the original commerce. Modern wallets handle the multi-token bookkeeping that this composition implies, which is why the kernel itself does not need to (and, by Theorem 4.7, must not). The composability posture is the architectural counterpart of the ideological-agnosticism claim above: the kernel makes the external surfaces composable; the graph names which externals it composes with. The companion implementation paper (§7) develops the formal *coordinator pattern* that makes such composition equilibrium-preserving; the present paper records only the architectural posture from which the pattern follows.

2 Related Work

Mutual-destruction escrows. The direct ancestor of FIGARO is the Safe Remote Purchase contract [Solidity Team, 2016], included in the Solidity documentation as a reference pattern: buyer and seller each lock 2× the item value, and only mutual agreement releases the funds. This two-party, single-order pattern is the seed from which FIGARO grew. BitHalo [BitHalo, 2014] independently explored mutual-destruction deposits for peer-to-peer Bitcoin trade.

The central problem was generalizing SRP beyond two parties. SRP uses symmetric 2× deposits with mutual agreement to release—both parties must cooperate. This works for a single exchange but cannot compose: in a twelve-seller delivery workflow, there is no natural "mutual agreement" among all participants. *Asymmetric bonding* (Theorem 3.1) solves this. By scaling each seller's bond with cumulative upstream value rather than fixing it per-trade, the mechanism creates a mesh of independently secured edges. Each edge carries its own Nash equilibrium (Theorem 4.3), and the mesh decomposes as finely as is operationally tolerable.

Buyer dominance then takes advantage of this mesh structure to make multi-party coordination resolvable from a single signature. Because a single party resolves the entire process tree atomically, coordination pressure propagates through the mesh without any management layer (Theorem 5.8).

State channels and payment networks. The Lightning Network [Poon and Dryja, 2016] and Raiden [Raiden, 2017] use time-locked deposits for off-chain payment. These protocols optimize for high-frequency bilateral payments and rely on timeouts for dispute resolution—precisely the escape hatch that FIGARO eliminates (Theorem 4.7). More critically, state channels do not address multi-party *coordination* (delivery, fulfillment, service composition); they address multi-hop *payment routing*.

On-chain dispute resolution. Kleros [Kleros, 2019] provides decentralized arbitration via Schelling-point juror selection. Aragon Court [Aragon, 2017] offers similar functionality within DAO governance. Both reintroduce a *discretionary human decision* into the resolution path, breaking the self-enforcing property (Theorem 4.7, Case 2).

Optimistic mechanisms. Optimistic rollups [Optimism, 2021] and optimistic oracles [UMA, 2020] use a challenge-response pattern: an assertion stands unless challenged within a timeout window. This requires liveness assumptions (the challenger must be online) and timeout calibration (too short enables fraud; too long delays finality). FIGARO has no timeout from the bonded state—capital remains locked indefinitely until the buyer resolves, making liveness irrelevant to security.

Mechanism design. The revelation principle [Myerson, 1981] guarantees that any achievable outcome can be implemented via a direct mechanism. FIGARO's insight is that the "direct mechanism" for multi-party coordination need not involve message passing, reporting, or arbitration at all: locked capital *is* the mechanism. The closest theoretical antecedent is the concept of *deposit games* [Asgaonkar and Krishnamachari, 2019], which analyze two-party security deposits. We extend deposit-game analysis to N-party DAGs with progressive collateralization.

Joint-liability microfinance. Group lending models—most prominently the Grameen Bank [Grameen Bank, 1999] and its formalizations by Ghatak [1999] and Stiglitz [1990]—show that joint liability combined with a social substrate can produce peer selection and peer monitoring outcomes. We return to these models in Section 6: FIGARO reproduces the same peer-enforcement outcome under strictly weaker assumptions, with no appeal to repeated interaction or local information.

3 The Mechanism

3.1 Notation and Setup

Consider two parties, a *buyer* B and a *seller* S, who wish to exchange a payment $P > 0$ (denominated in a unit of account) for goods or services. Both parties hold sufficient balances.

Definition 3.1 (Asymmetric Bonding)**.** An *asymmetric bond* for a transaction with payment P and cumulative value $G \geq P$ is a pair of deposits:

$$C_b = 2P \quad \text{(buyer bond)}, \tag{1}$$
$$C_s = 2G \quad \text{(seller bond)}. \tag{2}$$

For a root transaction (no upstream value), $G = P$, so both bonds equal $2P$. For sub-orders in a process tree, $G > P$, creating an asymmetry where the seller's bond exceeds the buyer's.

Definition 3.2 (Commitment)**.** A *commitment* is a tuple $\langle B, S, \tau, P, h, \sigma, \delta \rangle$ where B is the buyer identity, S the seller identity, τ the unit of account, P the payment, h a content hash of the off-chain agreement, σ a cryptographic salt, and δ a signature expiry deadline. Both parties sign the commitment off-chain.

Definition 3.3 (Process)**.** A *process* Π is a tuple $\langle \text{id}, B, \tau, G, n, \mathcal{O} \rangle$ where id is a deterministic identifier derived from the root commitment, B is the root buyer (constant across all orders), τ is the denomination (constant), G is a *monotonic accumulator* tracking total committed value, n is the count of active orders, and $\mathcal{O}$ is the set of active order identifiers.

3.2 Protocol Operations

The mechanism exposes exactly two state-changing operations.

Operation 1: `commit`. Given a commitment c signed by both B and S:

1. Verify both signatures.
2. If $c.\text{processId} = 0$, derive a new process identifier from the commitment digest and initialize the process with $G \leftarrow P$ and $n \leftarrow 1$.
3. Otherwise verify that $c.\text{processId}$ identifies an existing process, that signer B matches the process root buyer (*buyer dominance*), and that $c.\text{expectedCumulativeValue} = \Pi.G + P$.
4. Pull $C_b = 2P$ from B and $C_s = 2c.\text{expectedCumulativeValue}$ from S.
5. Create or extend the process accordingly, update the monotonic accumulator, and emit an evidence event that fully records the commitment.

Operation 2: `resolveProcess`**.** Given a process identifier and the complete list of active commitments:

1. Verify the caller identity equals $\Pi.B$ (buyer dominance).
2. Verify that the list length equals $\Pi.n$ (atomic resolution: *all* orders must be included).
3. For each order o_i with payment P_i and cumulative snapshot G_i:

$$\pi_{s,i} = 2G_i + P_i, \tag{3}$$
$$\pi_{b,i} = P_i. \tag{4}$$

4. Transfer $\pi_{s,i}$ to each seller, $\pi_{b,i}$ to the buyer.
5. Mark all orders as resolved.

Remark 3.4 (Conservation Law)*.* For each order, the total distributed equals total locked:

$$\pi_{s,i} + \pi_{b,i} = (2G_i + P_i) + P_i = 2G_i + 2P_i = C_{s,i} + C_{b,i}. \tag{5}$$

The escrow holds zero balance after resolution.

Remark 3.5 (No Escape Hatches)*.* There is no timeout, no cancel-after-commitment, no admin override, no governance vote, and no unilateral exit from the bonded state. Once both parties have committed, the only path to fund recovery is buyer-initiated resolution. This is not a temporary simplification; it is a *security requirement.* Theorem 4.7 proves that any additional exit path weakens the Nash equilibrium.

3.3 Net Payoffs

Definition 3.6 (Generalized Payout Functions)**.** For a bond multiplier $k \geq 1$, an order with payment $P > 0$ and cumulative value $G \geq P$, define the *settlement payout functions*:

$$\pi_s(k, G, P) = k \cdot G + P, \tag{6}$$
$$\pi_b(k, P) = (k-1) \cdot P. \tag{7}$$

These satisfy the conservation law $\pi_s + \pi_b = k \cdot G + k \cdot P = C_s + C_b$ for $C_s = k \cdot G$ and $C_b = k \cdot P$. The *net payoff* is the payout minus the deposited bond:

$$u_S^{\text{net}}(k) = \pi_s - C_s = P, \tag{8}$$
$$u_B^{\text{net}}(k) = \pi_b - C_b = -P. \tag{9}$$

Remark 3.7. For the mechanism's $k = 2$: $\pi_s = 2G + P$, $\pi_b = P$, $C_s = 2G$, $C_b = 2P$. The net payoffs ($+P$ for the seller, $-P$ for the buyer) are independent of k—the multiplier affects only the capital at risk, not the final value transfer.

4 Game-Theoretic Analysis

4.1 Two-Party Nash Equilibrium

Definition 4.1 (The Bonding Game)**.** The *bonding game* $\Gamma = (\{B, S\}, \{A_B, A_S\}, \{u_B, u_S\})$ is a one-shot, simultaneous-move, complete-information game:

- **Players**: buyer B, seller S.
- **Action sets**: $A_B = A_S = \{\mathrm{C}, \mathrm{D}\}$.

- **Interpretation**: C for S means delivering the agreed goods/services; C for B means calling `resolveProcess`. D means failing to perform.
- **Payoff functions** (net of bond deposits), for a root order with payment $P > 0$ and $G = P$:

$$u_B(a_B, a_S) = \begin{cases} -P & \text{if } a_B = \text{C} \text{ and } a_S = \text{C}, \\ -2P & \text{otherwise.} \end{cases} \tag{10}$$

$$u_S(a_B, a_S) = \begin{cases} +P & \text{if } a_B = \text{C} \text{ and } a_S = \text{C}, \\ -2G & \text{otherwise.} \end{cases} \tag{11}$$

The "otherwise" payoff is identical across all non-cooperative outcomes because any defection prevents resolution: bonds remain locked indefinitely, yielding a net loss equal to the full deposit.

Table 1: Payoff matrix for the bonding game Γ. Entry format: (u_B, u_S). Parameters: $P > 0$, $G \geq P > 0$.

	$a_B = \text{C}$	$a_B = \text{D}$
$a_S = \text{C}$	$(-P,\ +P)$	$(-2P,\ -2G)$
$a_S = \text{D}$	$(-2P,\ -2G)$	$(-2P,\ -2G)$

Definition 4.2 (Dominance)**.** Action a_i *weakly dominates* a_i' for player i if $u_i(a_i, a_{-i}) \geq u_i(a_i', a_{-i})$ for all a_{-i}, with strict inequality for at least one a_{-i}. If the inequality is strict for *all* a_{-i}, the dominance is *strict*.

Theorem 4.3 (Two-Party Nash Equilibrium)**.** *In the bonding game* Γ *of Theorem 4.1:*

(a) C *weakly dominates* D *for both the buyer and the seller.*

(b) (C, C) *is the unique Nash equilibrium.*

(c) (C, C) *is the unique Pareto-optimal outcome.*

Proof. **Part (a): Weak dominance.**
Buyer. We compare $u_B(\text{C}, a_S)$ vs. $u_B(\text{D}, a_S)$ for each seller action:

$$a_S = \text{C}: \quad u_B(\text{C}, \text{C}) = -P > -2P = u_B(\text{D}, \text{C}), \tag{12}$$
$$a_S = \text{D}: \quad u_B(\text{C}, \text{D}) = -2P = u_B(\text{D}, \text{D}). \tag{13}$$

Inequality (12) is strict (since $P > 0$); (13) is an equality. By Theorem 4.2, C weakly dominates D for B.
Seller. We compare $u_S(a_B, \text{C})$ vs. $u_S(a_B, \text{D})$ for each buyer action:

$$a_B = \text{C}: \quad u_S(\text{C}, \text{C}) = +P > -2G = u_S(\text{C}, \text{D}), \tag{14}$$
$$a_B = \text{D}: \quad u_S(\text{D}, \text{C}) = -2G = u_S(\text{D}, \text{D}). \tag{15}$$

Inequality (14) is strict (since $P > 0$ and $G > 0$ imply $P + 2G > 0$); (15) is an equality. C weakly dominates D for S.
Part (b): Unique Nash equilibrium. Since C weakly dominates D for both players, neither can profitably deviate from (C, C). We verify no other profile is a Nash equilibrium:

- (C, D): Seller deviates to C, gaining $P - (-2G) = P + 2G > 0$. Not a NE.
- (D, C): Buyer deviates to C, gaining $-P - (-2P) = P > 0$. Not a NE.

- (D, D): Buyer deviates to C with no change (both yield $-2P$). Seller deviates to C with no change (both yield $-2G$). This profile satisfies the Nash condition (no *profitable* deviation), but an iterated elimination of weakly dominated strategies removes D for both players, leaving (C, C) as the unique survivor.

Part (c): Pareto optimality. The payoff vector $(-P, +P)$ Pareto-dominates $(-2P, -2G)$ since $-P > -2P$ and $+P > -2G$. ☐

Remark 4.4 (On Weak vs. Strict Dominance). The dominance is weak, not strict, because when the counterparty defects, both actions yield the same locked-capital payoff. This is intrinsic to the mechanism: once one party defects, the outcome is fully determined regardless of the other's choice. The critical property is that (D, D) does not survive iterated elimination of weakly dominated strategies—the standard refinement for mechanism design [Myerson, 1981].

4.2 Sufficiency of the $2\times$ Multiplier

We now characterize the multiplier k in the generalized bonding game where $C_s = k{\cdot}G$, $C_b = k{\cdot}P$, and payouts are given by Theorem 3.6.

Lemma 4.5 (Buyer Indifference at $k = 1$)**.** *For $k = 1$, the buyer is indifferent between cooperating and defecting when the seller cooperates: $u_B(\mathrm{C},\mathrm{C}) = u_B(\mathrm{D},\mathrm{C}) = -P$.*

Proof. With $k = 1$: $C_b = P$ and $\pi_b = (k-1)P = 0$ (from (7)). Under our accounting convention (Theorem 3.6), the buyer's net payoff under cooperation is $u_B(\mathrm{C},\mathrm{C}) = \pi_b - C_b = 0 - P = -P$. Under defection, the buyer's bond remains locked: $u_B(\mathrm{D},\mathrm{C}) = -C_b = -P$. Since $u_B(\mathrm{C},\mathrm{C}) = u_B(\mathrm{D},\mathrm{C}) = -P$, the buyer has no strictly profitable deviation from D—the mechanism provides zero marginal incentive to resolve. ☐

Theorem 4.6 (Minimal Viable Bond Multiplier)**.** *Let $k \in \mathbb{Z}_{\geq 1}$ be the bond multiplier. Then:*

(a) For $k = 1$, C *does not weakly dominate* D *for the buyer: the buyer is indifferent when the seller cooperates.*

(b) For $k = 2$, C *weakly dominates* D *for both players (Theorem 4.3).*

(c) For $k > 2$, the equilibrium properties are identical to $k = 2$, but capital requirements increase by $(k-2)(G + P)$ per order with no improvement in dominance relations.

Thus $k = 2$ is the minimal integer multiplier that yields weak dominance of C *for both players, and is Pareto-optimal in capital efficiency among all valid multipliers.*

Proof. **Part (a).** See Theorem 4.5. With $k = 1$, the buyer recovers $\pi_b = 0$ upon resolution, having deposited $C_b = P$. The net monetary payoff from cooperating is $-P$; the net from defecting is also $-P$ (bond locked). The buyer cannot be motivated by the mechanism alone to resolve.

Part (b). With $k = 2$: $\pi_b = P$, $C_b = 2P$. Cooperation yields $u_B = -P$; defection yields $u_B = -2P$. Since $-P > -2P$ (as $P > 0$), the buyer strictly prefers cooperation when the seller cooperates. By Theorem 4.3(a), this establishes weak dominance.

Part (c). For $k \geq 2$: $u_B(\mathrm{C},\mathrm{C}) = -P$ and $u_B(\mathrm{D},\mathrm{C}) = -kP$. Since $-P > -kP$ for all $k \geq 2$, the dominance relation holds. The surplus capital locked, $(k-2)P$ for the buyer and $(k-2)G$ for the seller, earns zero return and is pure deadweight cost. ☐

4.3 Escape-Hatch Weakness

Theorem 4.7 (Escape-Hatch Weakness)**.** *Let Γ_E be the bonding game augmented with a unilateral exit action E available to player i, returning fraction $\alpha \in (0,1]$ of player i's bond. Then either:*

(a) C *no longer weakly dominates* D *for at least one player (the Nash equilibrium of Theorem 4.3 may not survive), or*

(b) *E requires a decision by a third party $J \notin \{B,S\}$ whose incentives are not constrained by the bond structure.*

Proof. We analyze three exhaustive sub-cases of unilateral exit.

Case 1: Timeout (automatic exit after duration T). The action space becomes $A_B = \{\mathrm{C}, \mathrm{D}, \mathrm{Wait}\}$ where Wait means the buyer does not resolve and collects $\alpha \cdot C_b$ after time T. The buyer's payoff under Wait against a cooperating seller is:

$$u_B(\mathrm{Wait}, \mathrm{C}) = -C_b + \alpha \cdot C_b = -2P(1-\alpha). \tag{16}$$

Compare with cooperation: $u_B(\mathrm{C}, \mathrm{C}) = -P$. The buyer prefers C to Wait iff:

$$-P > -2P(1-\alpha) \iff 2(1-\alpha) > 1 \iff \alpha < \tfrac{1}{2}. \tag{17}$$

For $\alpha \geq \frac{1}{2}$, Wait yields $u_B \geq -P = u_B(\mathrm{C}, \mathrm{C})$, so C no longer (weakly) dominates Wait. The buyer may rationally wait out the timeout instead of resolving. Even for $\alpha < \frac{1}{2}$, the defection cost drops from $2P$ to $2P(1-\alpha)$, weakening the deterrent. The seller, anticipating the timeout, faces a modified game where the buyer's commitment to resolve is no longer credible—the timeout provides an alternative path to partial capital recovery, undermining the "no escape" property that sustains the equilibrium.

Case 2: Arbitrator override. A third party J can invoke E on behalf of either player. Formally, this extends the game to three players $\{B, S, J\}$. The mechanism's self-enforcing property requires that the payoff structure *alone* determines the outcome. Introducing J makes the outcome contingent on J's action, and J is not bonded: J deposits nothing, risks nothing, and has no mechanism-constrained incentive to decide correctly. Whether J is an individual (arbitrator), an algorithm (oracle), or a collective (jury), the resolution path now depends on an agent outside the bond structure. This is condition (b).

Case 3: Governance vote. A special case of Case 2 where J is replaced by a set of voters $\{J_1, \ldots, J_m\}$ who collectively decide on E. This inherits all problems of Case 2, plus: (i) coordination failures among voters, (ii) plutocratic capture if voting power is token-weighted, (iii) voter apathy (rational ignorance). Each voter J_k is unbonded.

In all three cases, either C ceases to weakly dominate all alternatives (Case 1), or resolution depends on an unbonded external agent (Cases 2–3). □

Remark 4.8 (On Judicial Review and the Distinction from Cases 2–3)*.* The theorem's "unbonded external agent" condition targets actors whose decision is *constitutive* of the resolution path internal to the mechanism — arbitrators, juries, oracles, voters embedded in the bonding game itself. It does not target the external legal forum that may, on the evidence record, adjudicate a dispute *outside* the bonding game under doctrines of duress, frustration, impossibility, unconscionability, or public policy. Those forums are constrained by their own institutional bond structures (appellate review, professional sanction, doctrinal consistency) and operate on the bonded commitment as evidentiary input rather than as a resolution path within the mechanism. The distinction is elaborated in the companion evidence-and-law paper.

Corollary 4.9. *No timeout duration T and no recovery fraction $\alpha > 0$ can be added to the bonding game without weakening the buyer's incentive to cooperate. In particular, full recovery ($\alpha = 1$) reduces the game to a zero-cost option, making defection strictly dominant for the buyer.*

5 N-Party Process Trees

5.1 Progressive Collateralization

We now extend the two-party model to an arbitrary-length chain of sellers.

Definition 5.1 (Process Tree)**.** A *process tree* is a sequence of bonded orders $\langle o_1, o_2, \ldots, o_n \rangle$ sharing a common buyer B and denomination τ. Order o_i has local payment $P_i > 0$ and cumulative value:

$$G_i = \sum_{j=1}^{i} P_j. \tag{18}$$

The seller of o_i posts bond $C_{s,i} = 2G_i$; the buyer posts bond $C_{b,i} = 2P_i$.

Definition 5.2 (N-Party Bonding Game)**.** The *N-party bonding game* $\Gamma_N = (\{B, S_1, \ldots, S_n\}, \{A_i\}, \{u_i\})$ extends Γ to n sellers. Each seller S_i has action set $\{\mathrm{C}, \mathrm{D}\}$. The buyer has action set $\{\mathrm{C}, \mathrm{D}\}$ and chooses after observing each seller's realized action; we assume perfect monitoring, so B perfectly observes whether each S_i delivered, and plays the conditional strategy "resolve iff all sellers cooperated." The perfect-monitoring assumption corresponds to the deployed setting, in which the buyer receives either the agreed goods or not, and the off-chain observation is prior to the on-chain `resolveProcess` call. When monitoring is imperfect, the conditional strategy must be replaced by a signal-contingent resolution rule; we discuss this extension in Section 7. Payoffs under perfect monitoring:

$$u_{S_i}(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} +P_i & \text{if } a_B = \mathrm{C} \text{ and } a_{S_j} = \mathrm{C} \ \forall j, \\ -2G_i & \text{otherwise.} \end{cases} \tag{19}$$

The buyer's payoff is:

$$u_B(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} -\sum_{i=1}^{n} P_i & \text{if } a_B = \mathrm{C} \text{ and } a_{S_j} = \mathrm{C} \ \forall j, \\ -\sum_{i=1}^{n} 2P_i & \text{otherwise.} \end{cases} \tag{20}$$

Theorem 5.3 (N-Party Nash Equilibrium)**.** *In Γ_N, cooperation is a weakly dominant strategy for every seller S_i at every position $i \in \{1, \ldots, n\}$, and $(\mathrm{C}, \mathrm{C}, \ldots, \mathrm{C})$ is the unique outcome surviving iterated elimination of weakly dominated strategies.*

Proof. Fix an arbitrary seller S_i. Consider the two cases for the remaining players' joint action profile a_{-i}.

Case 1: All other sellers cooperate and buyer resolves. Then:

$$u_{S_i}(\ldots, \mathrm{C}_i, \ldots) = +P_i, \tag{21}$$
$$u_{S_i}(\ldots, \mathrm{D}_i, \ldots) = -2G_i. \tag{22}$$

Since $P_i > 0$ and $G_i > 0$, we have $P_i > -2G_i$, i.e., $P_i + 2G_i > 0$. The cooperation payoff strictly exceeds the defection payoff.

Case 2: At least one other seller defects, or buyer defects. Then resolution does not occur regardless of S_i's action:

$$u_{S_i}(\ldots, \mathrm{C}_i, \ldots) = u_{S_i}(\ldots, \mathrm{D}_i, \ldots) = -2G_i. \tag{23}$$

Payoffs are equal.

Combining both cases, C weakly dominates D for S_i, with strict inequality in Case 1. Since the choice of i was arbitrary, this holds for all n sellers. The buyer's dominance follows from Theorem 4.3(a), generalized: the buyer's payoff under cooperation ($-\sum P_i$) strictly exceeds the payoff under defection ($-2\sum P_i$) when all sellers cooperate.

Iterated elimination: in the first round, D is eliminated for all players (weakly dominated). The unique surviving profile is $(\mathrm{C}, \ldots, \mathrm{C})$. □

Remark 5.4 (Strength of the N-Party Result). Note that the seller's dominance condition, $P_i + 2G_i > 0$, holds with a margin that *grows* with chain depth (since $G_i = \sum_{j \le i} P_j$ is monotonic). The cooperation-defection payoff gap for S_i is:

$$\Delta_i = P_i - (-2G_i) = P_i + 2G_i = P_i + 2\sum_{j=1}^{i} P_j \ge 3P_i. \tag{24}$$

The gap is at least $3P_i$ (equality only at the root where $G_i = P_i$) and grows with every upstream payment. This is the formal basis of the "geometric coordination pressure" in Theorem 5.5.

Corollary 5.5 (Geometric Coordination Pressure)**.** *The risk-to-reward ratio for seller S_i is:*

$$\rho_i = \frac{C_{s,i}}{P_i} = \frac{2G_i}{P_i} = \frac{2\sum_{j=1}^{i} P_j}{P_i}. \tag{25}$$

For equal payments ($P_j = p$ for all j), $\rho_i = 2i$, growing linearly with chain depth. For typical value chains where early stages contribute the majority of value ($P_1 \gg P_n$), ρ_n grows super-linearly.

Example 5.6. Alice orders food (\$10) from Bob, who needs delivery (\$2) from Charlie:

Party	Role	P_i	G_i	$C_{s,i}$	ρ_i
Bob	Seller 1	\$10	\$10	\$20	2.0
Charlie	Seller 2	\$2	\$12	\$24	12.0

Charlie risks \$24 to earn \$2 ($\rho = 12$). Bob risks \$20 to earn \$10 ($\rho = 2$). The deeper participant faces amplified consequences.

5.2 Atomic Resolution and Seller Coordination

Definition 5.7 (Atomic Resolution)**.** Resolution is *atomic*: the buyer must resolve all active orders in a process simultaneously, or none. There is no partial resolution.

Proposition 5.8 (Endogenous Coordination Pressure)**.** *The N-party bonding game under atomic resolution induces a* weakest-link public goods game *among sellers, where each seller's payout depends on universal cooperation.*

Proof. Consider the sellers' subgame (fixing the buyer's strategy at: resolve iff all sellers cooperate). Each S_i has payoff u_{S_i} as in (19).

Define the *externality* imposed by S_k's defection on S_i ($i \neq k$):

$$\varepsilon_{k \to i} = u_{S_i}(\text{all C}) - u_{S_i}(S_k \text{ defects}) = P_i - (-2G_i) = P_i + 2G_i. \tag{26}$$

This is strictly positive for all i, k with $i \neq k$. Every defection imposes a loss on every other seller. The total externality of S_k's defection is:

$$\mathcal{E}_k = \sum_{i \neq k} \varepsilon_{k \to i} = \sum_{i \neq k} (P_i + 2G_i). \tag{27}$$

This structure satisfies the defining properties of a *weakest-link game* [Hirshleifer, 1983]:

1. The group outcome depends on the minimum individual effort (one defection collapses all payoffs to the non-cooperative level).
2. The cooperative outcome $(\text{C}, \dots, \text{C})$ is the unique Pareto-optimal Nash equilibrium of the subgame.

3. The game exhibits *strategic complementarity*: S_i's marginal return to cooperation is highest when all others cooperate.

In a repeated-interaction setting, this externality structure creates endogenous peer monitoring: S_i has a direct economic interest of magnitude $P_i + 2G_i$ in ensuring S_k cooperates. This incentivizes information acquisition about co-sellers, pressure application on underperformers, and reputation-based exclusion of known defectors—all without explicit communication protocols or governance mechanisms. $\square$

5.3 Self-Enforcing DAG Topology

Process trees can express arbitrary directed acyclic graph (DAG) topologies. A natural question: must the settlement layer validate parent-child relationships?

We treat the question in two settings, which answer it for different reasons.

Setting A: accumulator-strict (on-chain). If the settlement layer enforces the monotonic accumulator via a strict equality check $G'_i = \Pi.G_{\text{prev}} + P_i$, as the reference implementation does, then the strategy space for "report" is a singleton: any deviation — either under-reporting or over-reporting — causes the commit transaction to revert, and the order never enters the bonded state. The honesty property is therefore a direct consequence of the contract's revert semantics, not a game-theoretic derivation. We record this as an observation rather than a theorem: *in accumulator-strict settings, truth-telling is the only feasible action.*

Setting B: accumulator-soft (off-chain DAG or permissive layer). If the settlement layer permits over-reporting ($G'_i \geq G_i$), as would be the case in an off-chain DAG representation or a layer that bonds against declared rather than accumulated value, then honesty is not automatic and the equilibrium argument below applies.

Theorem 5.9 (Self-Enforcing DAG Honesty (Setting B)). *In the accumulator-soft setting, suppose the probability $q \in [0, 1]$ that the buyer resolves the process is monotonically non-increasing in S_i's reported cumulative value, and $q(G_i) > 0$ whenever the remaining sellers cooperate. Then truth-telling ($G'_i = G_i$) weakly dominates all feasible over-reports, with strict dominance whenever $q(G_i) < 1$.*

Proof. Write S_i's expected payoff as a function of the report G'_i and the buyer's resolution probability $q(G'_i)$:

$$\mathbb{E}[u_{S_i}(G'_i)] = q(G'_i) \cdot P_i - (1 - q(G'_i)) \cdot 2G'_i. \tag{28}$$

The first term is non-increasing in G'_i by the monotonicity assumption on q; the second term is strictly decreasing in G'_i (since $1 - q(G'_i) \geq 0$ and the coefficient is -2). The expectation is therefore maximized at the smallest feasible G'_i, namely G_i.

When $q(G_i) < 1$, the second term is strictly negative for $G'_i > G_i$, so the dominance is strict. $\square$

Remark 5.10 (On the monotonicity assumption). The assumption that q is non-increasing in G'_i is defensible in accumulator-soft settings because an inflated declared value imposes an enlarged bond on the buyer elsewhere in the process tree (via progressive collateralization), which reduces the buyer's willingness to extend the tree; a rational buyer will treat an inflated G'_i as evidence of misreporting and may decline to resolve. The assumption does *not* hold for buyers who are insensitive to reported value — a corner case in which q is constant in G'_i and the theorem reduces to the accumulator-strict observation.

6 Reduction from Microfinance Mechanisms

The coordination pressure of Theorem 5.8 is operationally analogous to the peer-enforcement dynamics documented in the microfinance literature on group lending—most prominently the Grameen Bank [Grameen Bank, 1999] and its formalizations by Ghatak [1999] and Stiglitz [1990]. The analogy is frequently drawn informally, but the precise relationship between the bonded commitment mechanism and the microfinance models has not, to our knowledge, been stated formally. We make it explicit here because it identifies what FIGARO actually contributes relative to the prior literature on peer-enforced cooperation.

The microfinance mechanisms. Two canonical models frame group lending. Ghatak [1999] proves *peer selection*: under joint-liability contracting, borrowers with private type information sort assortatively, attenuating adverse selection. The result requires a heterogeneous population, local information among borrowers, and voluntary group formation. Stiglitz [1990] proves *peer monitoring*: joint liability combined with repeated social interaction lets group members observe and punish co-borrower deviation, attenuating moral hazard. The result requires repeated interaction, observability of co-borrower actions, and an exogenous punishment technology (social sanction, denial of future credit). In both cases the peer-pressure outcome is an *emergent* property of a contract (joint liability) embedded in a social substrate (repeated interaction, local information). The contract alone is insufficient; the substrate does enforcement work.

Proposition 6.1 (Assumption Reduction on Cooperation Pressure)**.** *The* cooperation-pressure component *of the equilibrium common to Ghatak [1999] and Stiglitz [1990]—cooperative equilibrium selection driven by co-participant externalities—is obtained by the bonded commitment mechanism under strictly weaker assumptions. Specifically, FIGARO requires none of the following: (i) repeated interaction, (ii) local information among sellers, (iii) a punishment technology exogenous to the contract, nor (iv) joint-liability contracting.*

Proof. Theorem 5.3 establishes cooperation as the unique profile surviving iterated elimination of weakly dominated strategies in a one-shot game. The argument uses only the on-chain payoff function and makes no appeal to continuation value, trigger strategies, or folk-theorem constructions; (i) is not required.

The externality $\varepsilon_{k\to i} = P_i + 2G_i$ (Theorem 5.8) is computed from quantities enforced by the settlement layer's monotonic accumulator, independent of any seller's information about co-sellers' types, actions, or identities; (ii) is not required.

The punishment for a co-seller's defection is the loss of S_i's own bond $C_{s,i} = 2G_i$, imposed directly by the contract through non-resolution. No external punishment technology—social sanction, reputational downgrade, or denial of future credit—is invoked; (iii) is not required.

Finally, each $C_{s,i}$ is posted individually by S_i against S_i's own cumulative-value snapshot, not against the group's aggregate obligation. Coupling across sellers arises from the buyer's atomic-resolution constraint (Theorem 5.7), not from the bond structure; (iv) is not required. □

What is *not* reproduced. The proposition is scoped deliberately. Group-lending models derive two distinct equilibrium effects, and FIGARO reproduces only the first.

- **Cooperation under externality coupling (reproduced).** Once a group is formed and joint liability is in force, each member has incentive to perform because a co-member's failure imposes direct loss. The bonded mechanism reproduces this effect under strictly weaker assumptions as shown above.
- **Peer selection under private type information (not reproduced).** Ghatak [1999] shows that joint-liability contracts induce *assortative matching* among borrowers of different unobservable risk types; this is an adverse-selection result and requires both a heterogeneous type space and a group formation stage. FIGARO has no type-screening mechanism

and no group-formation stage that selects on unobservable seller quality. Adverse-selection screening must be obtained by some mechanism outside the bonded commitment (e.g., attested credentials, operator registries).

- **Peer monitoring under principal information asymmetry (not directly reproduced).** Stiglitz [1990] treats a setting where deliverable quality is unobservable to the principal (lender) but observable to peer borrowers. The equilibrium leverages the principal–peer information asymmetry. FIGARO's bonded commitment assumes the buyer can determine whether to resolve; it sidesteps rather than solves the buyer-side moral hazard. When the buyer's performance observation is itself noisy or delayed, the perfect-monitoring assumption in Theorem 5.2 becomes load-bearing and the cooperation-pressure reproduction depends on how that noise is handled at the mechanism layer.

The honest summary of Theorem 6.1 is therefore that it reproduces one of the two classical group-lending equilibrium effects, under weaker assumptions, for a correspondingly narrower conclusion. The other effect — assortative screening on unobservable types — requires additional mechanism above the bonded kernel.

What this repositions. The reduction does not correct the microfinance theorems—they are sound under their assumptions—but it reframes them. Repeated interaction and local information were necessary for group lending because the enforcement primitive available in that setting was reputation within a social network. Once the settlement layer admits bilaterally signed, individually collateralized commitments with atomic resolution, the same mechanism outcome obtains among mutual strangers, in a single shot, with no social substrate. Repeated interaction and local information are thus not necessary conditions for endogenous peer monitoring; they are features of one particular settlement environment in which the mechanism was historically deployed. The weakest-link subgame (Theorem 5.8), the geometric risk amplification (Theorem 5.5), and the one-shot dominance argument (Theorem 5.3) together give a purely mechanism-level construction of an effect the microfinance literature attributes to contract design combined with social embedding.

7 Conclusion and Open Questions

We have presented FIGARO, a settlement mechanism that achieves self-enforcing multi-party coordination through asymmetric bonding. The mechanism is minimal—no timeout, no governance, no escape hatches—and the game-theoretic properties are strong: cooperation is weakly dominant at every position in an N-party process tree, and the cooperative profile is the unique outcome surviving iterated elimination of weakly dominated strategies. The mechanism reproduces the peer-enforcement effect of joint-liability microfinance under strictly weaker assumptions, with no appeal to repeated interaction, local information, or social sanction.

Implementation and verification. A reference implementation as a Solidity smart contract, together with the formal-verification stack (TLA$^+$, symbolic execution, property-based fuzzing, specification checking) and the threat model that accompanies it, is presented in the companion implementation paper. That paper also develops the *coordinator pattern*: sufficient conditions under which an external mechanism composed with the kernel preserves the bonding equilibrium, and the corresponding three-layer enforcement architecture (economic / coordination / evidence).

Organizational implications. The implications of a fixed-cost enforcement primitive for the boundary of the firm in the sense of Coase [1937]—and specifically for the demotion of the firm from privileged organizational container to one pattern among many (alongside transaction-scoped assemblies, treasury communities, peer networks, and agent coalitions) when the subordination overlay is no longer enforcement-driven—are developed in the companion institutional-economics paper.

Focal-point coordination among participants. The companion FIG token paper develops FIG, a focal-point coordination token in the sense of Schelling [1960], designed to support convergence among participants aligned with the bonded primitive without any kernel-level interaction. FIG is an instance of the compositional discipline sketched above: a sibling mechanism that does not call `commit` or `resolveProcess`, holds no kernel bonds, and modifies no kernel state, thereby preserving the bonding equilibrium trivially. The pattern — a native token whose value derives from its focality, underpinned by a distinct real-economic quantity at a distinct layer of the economic stack — has precedent in Nakamoto [2008] at the money layer (underpinned by cost of production) and Buterin [2014] at the programmable-settlement layer (underpinned by transaction demand). FIG extends the pattern to the composable-institution layer, underpinned by *trade* — the value-added produced when parties exchange goods, services, or coordination through the bonded primitive.

The accounting byproduct. A consequence of the conservation law (Theorem 3.4) worth flagging: the kernel is at all times in custodial balance, and a process is, in the precise accounting sense, a self-closing ledger period. The companion accounting paper develops the implications for the bookkeeping apparatus that grew up around double-entry [Pacioli, 1494] once that apparatus is made optional by protocol-level constitution of the records.

Open theoretical questions. The mechanism leaves several theoretical questions open. First, Theorem 5.3 bounds rational-actor equilibria; the behavior of the mechanism under bounded rationality, incomplete information about co-sellers' types, or non-pecuniary preferences (spite, fairness) is not characterized. Second, the mechanism is presented for sequential process trees with a single denomination; extending the dominance result to richer DAG topologies with cross-process composition, or to multi-currency processes (which require an external rate of substitution and so re-introduce a discretionary actor), remains open. Third, the relationship between the $2\times$ multiplier and the opportunity cost of locked capital warrants closer treatment: the present analysis shows that $k = 2$ is sufficient at zero opportunity cost and remains sufficient for $rt > 0$, but does not characterize the rate regime under which a richer dynamic mechanism (e.g., interest-bearing bonds, time-varying multipliers) would strictly Pareto-dominate the static $k = 2$ design.

A scope note on voluntary participation. The equilibrium results of this paper hold under the assumption that both parties are legally and materially free to enter the bonding game — i.e., that entering is a choice and that capital sufficient to post the required bond is available. For participants whose only realistic income path runs through bonded assemblies, or who lack access to the bond-denomination currency, the "voluntary" framing is itself a variable whose treatment lies outside mechanism design. The distributional and labor-law consequences of this scope are developed in the companion institutional-economics and labor-law papers; the present paper establishes equilibrium properties conditional on voluntary participation, not the conditions under which participation is voluntary in any substantive sense.

Acknowledgments

I owe Vlad Zamfir a particular debt: his work in mechanism design and coordination economics gave me the conceptual vocabulary to recognize Buterin's Safe Remote Purchase contract not

merely as an escrow pattern but as a *coordination mechanism*—one that could, in principle, scale beyond two parties. Without that lens, the generalization to asymmetric bonding and N-party process trees would not have happened. I also want to acknowledge Vitalik Buterin for the Safe Remote Purchase primitive itself, Fabrizio Romano Genovese for his contributions to FIGARO, and the Coalition of Automated Legal Applications (COALA).

References

Cuende, L. and Izquierdo, J. Aragon Network: A Decentralized Infrastructure for Value Exchange. Aragon White Paper, 2017. `https://github.com/aragon/whitepaper`

Asgaonkar, A. and Krishnamachari, B. Solving the Buyer and Seller's Dilemma: A Dual-Deposit Escrow Smart Contract for Provably Cheat-Proof Delivery and Payment for a Digital Good. In *Proc. IEEE International Conference on Blockchain and Cryptocurrency*, 2019.

Dwyer, D. BitHalo: Decentralized Contracts and Escrow Without Trust. BitHalo White Paper, 2014.

Solidity Team. Safe Remote Purchase. In *Solidity Documentation — Solidity by Example*, 2016. `https://docs.soliditylang.org/en/latest/solidity-by-example.html#safe-remote-purchase`

Coase, R. H. The Nature of the Firm. *Economica*, 4(16):386–405, 1937.

Pacioli, L. *Summa de Arithmetica, Geometria, Proportioni et Proportionalita.* Paganinus de Paganinis, Venice, 1494.

Buterin, V. Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform. White paper, 2014. `https://ethereum.org/en/whitepaper/`

Nakamoto, S. Bitcoin: A Peer-to-Peer Electronic Cash System. White paper, 2008. `https://bitcoin.org/bitcoin.pdf`

Schelling, T. C. *The Strategy of Conflict.* Harvard University Press, Cambridge, MA, 1960.

Ghatak, M. Group lending, local information and peer selection. *Journal of Development Economics*, 60(1):27–50, 1999.

Yunus, M. The Grameen Bank. *Scientific American*, 281(5):114–119, 1999.

Hirshleifer, J. From weakest-link to best-shot: The voluntary provision of public goods. *Public Choice*, 41(3):371–386, 1983.

Ast, F. and Lesaege, C. Kleros: A Protocol for a Decentralized Justice System. Kleros White Paper, 2019. `https://kleros.io/yellowpaper.pdf`

Myerson, R. B. Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73, 1981.

Optimism Foundation. Optimistic Rollups. Optimism Technical Documentation, 2021. `https://community.optimism.io/docs/protocol/`

Poon, J. and Dryja, T. The Bitcoin Lightning Network: Scalable Off-Chain Instant Payments. Technical report, 2016.

Raiden Network. Raiden Network: Fast, Cheap, Scalable Token Transfers for Ethereum. Raiden White Paper, 2017. `https://raiden.network/`

Stiglitz, J. E. Peer monitoring and credit markets. *World Bank Economic Review*, 4(3):351–366, 1990.

UMA Project. The Optimistic Oracle. UMA Technical Documentation, 2020. https://docs.uma.xyz/